

Neural Network-Based Formation Control With Target Tracking for Second-Order Nonlinear Multiagent Systems

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This article proposes a distance-based formation control and target tracking for multiagent systems, where agents are modeled using second-order nonlinear systems in the presence of disturbance. By applying a rigid graph theory, we developed a neural network (NN)-based backstepping controller to address the distance-based formation control problem of nonlinear multiagent systems. To compensate for the unknown nonlinearity in the system dynamics, the radial basis function NN was used where the NN tuning law was derived based on Lyapunov stability theory. We rigorously proved the uniform ultimate boundedness of the formation distance error and NN weights' norm estimation error. Finally, using simulation results, we demonstrated the proposed method's performance on the second-order, nonlinear multiagent systems. To provide further evaluation, we compared the proposed distance-based method and existing displacement-based methods.

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I. INTRODUCTION

Multiagent systems have found applications to many different areas over the past two decades, including guidance and navigation of autonomous unmanned aerial vehicles (UAVs) deployed to perform surveillance and exploration [1], connected vehicles [2], traffic flow control [3], sensors monitoring [4], etc. The formation control systems from group behaviors in nature, e.g., school of fish, flock of birds, pack of wolves, and others. A general categorization of formation control includes position-based control, displacement-based control, and distance-based control [5]. Among these approaches, distance-based formation control requires less sensing capability in comparison with other methods because agents control only the interagent distances to achieve the predefined desired formation. Agents can sense the relative positions of their adjacent agents by utilizing their on-board sensors for control within the local coordinate frames [5]. In addition, this method requires a high number of interactions among agents, which plays a crucial role in the formation control of multiagent systems. In the displacement-based methods, the control goal is to achieve the desired displacement among neighboring agents, where each agent measures its states with respect to a global coordinate system. The consensus-based formation control can be categorized as a displacement-based control, where the desired relative displacements of agents are set to zero [5]. The leader–follower approach can be found in [6] and [7], and a combination of the leader–follower approach and distance-based formation control in [8].

The relation among agents can be modeled by directed or undirected graphs. Most developed control methods focus on the undirected graph topology for single-integrator multiagent systems [9]–[12]. However, in recent studies, double-integrator multiagent systems became more commonly used due to their practical applications. Distance-based formation control of double-integrator systems with undirected graph topology was studied in [1], [13], and [14]. Graph theory, and in particular graph rigidity, is used in distance-based control, where each agent measures distances from its neighboring agents instead of their exact locations in the global coordinate frame [15]. Ramazani *et al.* [1], studied single- and double-integrator multiagent systems over undirected graphs. Directed graph formations of single- and double-integrator systems through distance-based control were studied in [16], where leaders can sense their positions in a global reference framework, while the followers can only sense their neighbors' relative positions with regard to their own local reference frames.

Tian *et al.* [17], considered double-integrator dynamics of agents for the leader–follower consensus problem. Double-integrator distance-based formation control was studied in [18], where a discrete-time control algorithm was designed to achieve velocity consensus. Related to our work, Wang *et al.* [19] used the backstepping design to solve the problem of formation control for double-integrator agents. Distance-based formation control for single-integrator dynamics was addressed in [20], where

collision avoidance and desired formation are achieved using the angular information between the agents. Babazadeh *et al.* [21] addressed the distance-based formation control for single- and double-integrator dynamics with energy constraints, where normalized rigidity is used to asymptotically minimize energy consumption.

Neural network (NN) based control methods have widely been used in literature to approximate unknown nonlinearities of nonlinear multiagent systems [22]–[28]. In [22], an adaptive NN control method for a class of first-order nonlinear multiagent systems with state time-delay was proposed, where an undirected graph models the communication among the agents.

In [23], a self-structuring adaptive NN controller for multiagent systems with second-order nonlinear dynamics was presented where the control goal is to achieve a leader-following formation over a directed connected graph. In [24], a cooperative adaptive consensus controller was designed based on the NN approximation method for a set of second-order nonlinear multiagent systems. In [25], particle tracking formation of second-order multiagent systems using an adaptive NN control with multiple leaders was presented. In [26], a consensus control based on graph Laplacian for a second-order nonlinear multiagent system was presented. The Frobenius norm of the NN weights is tuned, which reduces the computation complexity of the proposed approach. In [27], an adaptive NN-based distributed containment controller was designed to reach a leader-follower synchronization, where an NN is utilized to approximate the uncertain dynamics of a second-order nonlinear multiagent system.

In a preliminary, conference version of this article [29], we presented the formation control for a class of nonlinear systems in Brunovsky form. Compared with the article presented in [29], in this article, we study the input-affine, nonlinear multiagent systems, where the objective is to keep agents, modeled by second-order nonlinear systems, in prespecified distances from each other and to track a target within the bounded trajectory. We also present the proof that the tracking error is uniform ultimate boundedness (UUB) and demonstrate extensive simulation results in order to validate and verify the efficiency of the proposed method. We introduce a performance index to demonstrate advantages of the proposed method in comparison with the displacement-based control methods.

A bounded time-delay and disturbance rejection for the distance-based formation control for nonlinear multiagent systems appear in [30]. To compensate for the time-delayed function, there is a time-dependent control gain, which is different from the control design here. The disturbance dynamics is different compared to this article including the assumptions for the boundedness of the disturbance. A normalized rigidity matrix was used in [30], where an upper bound for the minimum singular value of the normalized rigidity matrix has been introduced and utilized in controller design.

To achieve the control objectives, we propose a control design for the nonlinear multiagent systems using the

backstepping control technique. The unknown nonlinear part of the system dynamics is compensated using an RBFNN similar to the article presented in [26], where authors used the norm of NN weights matrix. The leader follows the target while the rest of the agents maintain the desired formation and follow the leader [31]. Our results are based on an infinitesimally rigid graph, while displacement-based methods usually rely on the existence of a spanning tree [5].

The novelties are a rigorous analysis and a proof of formation stability control using NNs for second-order nonlinear multiagent systems in the presence of disturbance. The main contributions of this article are as follows.

- 1) Compared with existing distance-based formation control methods [1], [8]–[13], [20], which considered single- or double-integrator dynamics for agents, we assume that agents are modeled as the second-order nonlinear systems with unknown nonlinearity and disturbance. We provided a rigorous Lyapunov stability analysis for the formation control and target tracking for such a system. We extended the results in [29] and included input-affine, nonlinear systems with unknown nonlinearities.
- 2) In contrast with existing methods [22]–[26], [32] that used Laplacian matrix to model relations among the agents and address the displacement-based formation problem, we applied the *rigid graph theory* to address the problem of target tracking and distance-based formation control for input-affine, nonlinear multiagent systems.

The rest of this article is organized as follows. In Section II, the problem formulation and required background of rigid graph theory are given. In Section III, the design of a target tracking backstepping controller based on RBFNN using the Lyapunov technique for a second-order nonlinear multiagent system is introduced. In addition, the stability analysis and performance of the multiagent systems are presented. In Section IV, the numerical simulation results are presented, and they are compared with two different displacement-based methods to illustrate the effectiveness of the proposed method. Finally, Section V concludes this article.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Preliminaries

The relation among agents is described by an undirected graph that is denoted by $G = (V, E)$, with $V = \{v_1, \dots, v_N\}$ set of vertices and $E \subset V \times V$ set of edges. With $|V| = N$ and $|E|$, we indicate the number of vertices and edges, respectively. A pair of $F = (G, p)$ is known as a framework where $p = (p_1, \dots, p_N)$ and $p_i \in \mathbb{R}^d$, $d \in \{2, 3\}$ is the position of vertex i in 2-D or 3-D space. Related to any arbitrary ordering of edges in E , let us define the edge function of $\phi : \mathbb{R}^{dN} \rightarrow \mathbb{R}^{|E|}$ as follows:

$$\phi_G(p) = (\dots, \|p_i - p_j\|^2, \dots), (i, j) \in E \quad (1)$$

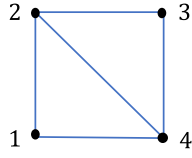


Fig. 1. Graph model of a system with four agents.

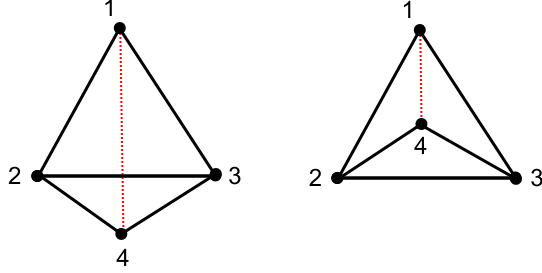


Fig. 2. Noncongruent framework due to flip ambiguity of vertex 4.

where $\|\cdot\|$ denotes the Euclidean norm. The rigidity matrix $R_p : \mathbb{R}^{dN} \rightarrow \mathbb{R}^{|E| \times dN}$ for the framework (G, p) is defined as

$$R_p = \frac{1}{2} \frac{\partial \phi_G(p)}{\partial p}. \quad (2)$$

For instance, consider an undirected graph shown in Fig. 1. The corresponding rigidity matrix R_p is given by

$$R_p = \begin{bmatrix} \tilde{p}_{12}^T & \tilde{p}_{21}^T & 0 & 0 \\ 0 & \tilde{p}_{23}^T & \tilde{p}_{32}^T & 0 \\ 0 & \tilde{p}_{24}^T & 0 & \tilde{p}_{42}^T \\ \tilde{p}_{14}^T & 0 & 0 & \tilde{p}_{41}^T \\ 0 & 0 & \tilde{p}_{34}^T & \tilde{p}_{43}^T \end{bmatrix} \quad (3)$$

where $\tilde{p}_{ij}^T = p_i^T - p_j^T = [x_i - x_j, y_i - y_j]$, and x_i, y_i denote position of vertex i in x, y coordinates, respectively. Without the loss of generality, the rigidity matrix can also be defined in 3-D [1].

LEMMA 1 (SEE [14]) Let framework (G, p) be given in 2-D. For any vector $v_r \in \mathbb{R}^2$, an N -size vector of ones denoted by $\mathbf{1}_N$, and an $|E|$ -size vector of zeros denoted by $\mathbf{0}_{|E|}$, the following identity holds:

$$R_p(\mathbf{1}_N \otimes v_r) = \mathbf{0}_{|E|} \quad (4)$$

where R_p is the corresponding rigidity matrix, and $\otimes$ denotes a Kronecker product.

Two frameworks (G, p) and (G, h) are equivalent, if $\phi_G(p) = \phi_G(h)$, and congruent if $\|p_i - p_j\| = \|h_i - h_j\|$ for all $i, j \in V$, [33], [34]. Two noncongruent equivalent frameworks are called ambiguous. Fig. 2 shows that the flip ambiguity of vertex 4 causes the frame to be noncongruent.

DEFINITION 1 (SEE [35]) Consider two frameworks $F = (G, p)$ and $F_1 = (G_1, q)$. These two frameworks, with $G = (V, E)$ and $G_1 = (V_1, E_1)$, are said to be isomorphic if they have similar vertex and edge sets, where there exists a bijection $f_1 : V \rightarrow V_1$ such that $v_i v_j \in E$ if and only if $f_1(v_i) f_1(v_j) \in E_1$.

Isomorphic frameworks to a framework F are denoted by $\text{Iso}(F)$ [36]. For a framework of F , the notation $\text{Amb}(F)$ denotes the set of all ambiguities [36].

The number of edges is important in determining the rigidity of a formation, as well as the distribution of the edges among the graph vertices. Laman's theorem relates rigidity and the number of edges and vertices in 2-D.

THEOREM 1 (SEE [37]) An undirected graph $G(V, E)$ in 2-D is rigid, if and only if there exists a subgraph $G_1 = (V, E_1)$, where $E_1 \subset E$ with $|E_1| = 2|V| - 3$ such that for any $V_1 \subset V$, the associated induced subgraph $G_2 = (V_1, E_2)$ of G_1 with $E_2 \subset E_1$, satisfies $|E_2| \leq 2|V_1| - 3$.

A minimally rigid graph in 2-D and 3-D has $|E| = 2N - 3$ and $|E| = 3N - 6$, respectively [37]. An interested reader is invited to see [38].

LEMMA 2 (SEE [39]) A framework with $N > 2$ in 2-D is infinitesimally rigid, if and only if $\text{rank}(R_p) = 2N - 3$.

DEFINITION 2 The neighborhood N_i of the vertex v_i is the set of all edges adjacent to v_i

$$N_i = \{v_j \in V \mid v_i v_j \in E\}. \quad (5)$$

DEFINITION 3 Time-dependent vector function $y(t)$ is said to be UUB [40] if there exists a compact set $\mathcal{S} \subset \mathbb{R}^N$ such that for all $y(t_0) \in \mathcal{S}$, there exist a bound B_S and a time $T_f(B_S, y(t_0))$, such that $\|y(t)\| \leq B_S$ for $\forall t \geq t_0 + T_f$.

The Frobenius norm of any matrix Q is $\|Q\|_F = \sqrt{\text{tr}(Q^T Q)}$, and $\text{tr}(\cdot)$ denotes the trace of a matrix.

B. Problem Formulation

Consider a nonlinear multiagent dynamical system with agents modeled by

$$\begin{aligned} \dot{p}_i &= v_i \\ \dot{v}_i &= f_i(p_i, v_i) + g_i(p_i, v_i)u_i(t) + w_i(t) \\ i &= 1, 2, \dots, N \end{aligned} \quad (6)$$

where the vectors $p_i \in \mathbb{R}^2$ and $v_i \in \mathbb{R}^2$ represent the position and velocity of each agent, respectively. Let us define $\tilde{\mathbf{x}}_i = [p_i^T, v_i^T]^T \in \mathbb{R}^4$. The nonlinear function $f_i(\tilde{\mathbf{x}}_i) \in \mathbb{R}^2$ is the unknown function and considered to be continuously differentiable, $g_i(\tilde{\mathbf{x}}_i) \in \mathbb{R}^{2 \times 2}$ is the input gain matrix function, $u_i \in \mathbb{R}^2$ is the control input, $w_i \in \mathbb{R}^2$ is a disturbance affecting each agent, and N represents the number of the agents.

The overall dynamics of the multiagent system is given by

$$\begin{aligned} \dot{\mathbf{x}}_1 &= \mathbf{x}_2 \\ \dot{\mathbf{x}}_2 &= f(\mathbf{x}_1, \mathbf{x}_2) + g(\mathbf{x}_1, \mathbf{x}_2)u + w \end{aligned} \quad (7)$$

where $\mathbf{x}_1 = [p_1^T, p_2^T, \dots, p_N^T]^T \in \mathbb{R}^{2N}$ and $\mathbf{x}_2 = [v_1^T, v_2^T, \dots, v_N^T]^T \in \mathbb{R}^{2N}$ are defined as overall positions and velocities of all agents, respectively. The vector $\mathbf{x} = [\mathbf{x}_1^T, \mathbf{x}_2^T]^T$, $f(\mathbf{x}) = [f_1^T(\tilde{\mathbf{x}}_1), f_2^T(\tilde{\mathbf{x}}_2), \dots, f_N^T(\tilde{\mathbf{x}}_N)]^T \in \mathbb{R}^{2N}$ is the nonlinear vector function of the whole system, $g(\mathbf{x}) =$

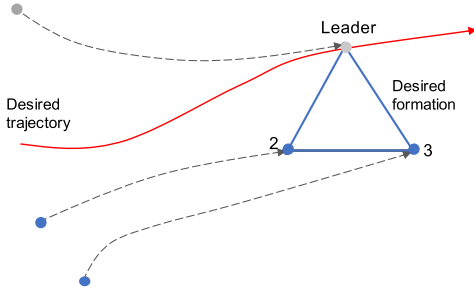


Fig. 3. Target tracking and formation control problems.

$\text{diag}(g_1(\tilde{\mathbf{x}}_1), g_2(\tilde{\mathbf{x}}_2), \dots, g_N(\tilde{\mathbf{x}}_N)) \in \mathbb{R}^{2N \times 2N}$, the control input vector is $u = [u_1^T, u_2^T, \dots, u_N^T]^T \in \mathbb{R}^{2N}$, and the disturbance vector is $w = [w_1^T, w_2^T, \dots, w_N^T]^T \in \mathbb{R}^{2N}$.

We establish some standard assumptions [24], [41].

ASSUMPTION 1 The unknown matrix $g_i(\tilde{\mathbf{x}}_i)$ is a positive-definite, symmetric matrix, with eigenvalues satisfying $0 < g_i \leq \lambda_1(g_i(\tilde{\mathbf{x}}_i)) \leq \lambda_2(g_i(\tilde{\mathbf{x}}_i)) < \infty$, and g_i being a constant lower bound of $g_i(\tilde{\mathbf{x}}_i)$.

Assumption 1 is needed for the controllability of system (6), which is a standard assumption in most NN-based controller designs [26], [41]–[43]. A large class of nonlinear systems described by Euler–Lagrange formulation satisfies this assumption [40], [44]. Let us define $g = \min(g_i)$.

ASSUMPTION 2 Disturbance $w_i(t)$ is unknown and bounded with a fix bound, i.e., $\|w_i\| \leq w_M$.

The objective of the control law is to ensure that distance between neighboring agents i and j to converge to the desired distance d_{ij}

$$\|p_i - p_j\| \rightarrow d_{ij} \text{ as } t \rightarrow \infty, (i, j) \in E. \quad (8)$$

ASSUMPTION 3 Desired distances between agents are bounded by $\|d_{ij}\| < D \forall (i, j) \in E$, with a fixed bound D .

Consider points $p_1, p_2 \in \mathbb{R}^2$ and a set $\mathcal{S}_p$. The distance between point p_1 and the set $\mathcal{S}_p$ is defined as $d(p_1, \mathcal{S}_p) \triangleq \inf_{p_2 \in \mathcal{S}_p} \|p_1 - p_2\|$. Let the initial conditions belong to the set $\mathcal{S}$ given by

$$\mathcal{S} = \left\{ \mathbf{x}_1 \in \mathbb{R}^{2N} \mid \sum_{(i,j) \in E^*} (\|p_{ij}\|^2 - d_{ij}^2)^2 \leq \delta \right. \\ \left. d(\mathbf{x}_1, \text{Iso}(F^*)) < d(\mathbf{x}_1, \text{Amb}(F^*)) \right\} \quad (9)$$

where δ is a positive constant [31], [36], [45]. For initial conditions belong to $\mathcal{S}$, the convergence of the formation to $\text{Iso}(F^*)$ and avoidance of getting stuck in $\text{Amb}(F^*)$ are guaranteed [36]. While the precise determination of δ is difficult when the number of agents increases [31], the sufficient condition to satisfy the inequality is given in [1]. The interested reader is invited to see [1], [31], [45], and references therein.

For target tracking, we consider the first agent to be the leader that tracks the target, while remaining agents follow the leader and retain the desired formation (see Fig. 3).

Let us define the tracking error between the leader and the target as [31]

$$e_r = p_r - p_1 \quad (10)$$

$e_r(t) = [e_{r1}, e_{r2}]^T$, and components e_{r1} and e_{r2} represent the leader's tracking errors in the x and y directions, respectively. Vectors $p_r, p_1 \in \mathbb{R}^2$ denote positions of the target and the leader, respectively, while $v_r \triangleq \dot{p}_r$ is the target (reference) velocity.

ASSUMPTION 4 The variables $p_r, \dot{p}_r, \ddot{p}_r$ are bounded.

Let us define a compact set [30] such that

$$\Omega_T = \{p_r, \dot{p}_r, \ddot{p}_r \mid \|p_r\| \leq \bar{P}_r, \|\dot{p}_r\| \leq \bar{V}_r, \|\ddot{p}_r\| \leq \bar{A}_r\} \quad (11)$$

where $\bar{P}_r, \bar{V}_r$, and $\bar{A}_r$ are the positive constants.

ASSUMPTION 5 The signals $p_r - p_1, \dot{p}_r - \dot{p}_1$, as well as v_r , are known signals and can be broadcasted to the followers [31].

REMARK 1 We assume that the leader can approximate the position and velocity of the moving target using, for example, the proposed technique in [46], where a moving target is detected, and its 2-D velocity is estimated through a moving target indication method [47] based on back projection and velocity synthetic aperture radar [46]. Moreover, the leader can distribute the relative position and velocity of the target to the followers [30], [31].

C. Radial Basis Function Neural Networks (RBFNNs)

We use an NN to compensate for the unknown nonlinearity in the system. Additional details on NNs can be found in [40]. We assume that the unknown nonlinearity in (6) is smooth; therefore, it can be approximated on a compact set Ω_i as

$$f_i(z) = W_i^T \varphi_i(z_i) + \epsilon_i(z_i) \quad (12)$$

where $z_i \in \Omega_i \subset \mathbb{R}^4$, and $\varphi_i(z_i) \in \mathbb{R}^{\eta_i}$ is a known smooth function, η_i is the number of neurons, $W_i \in \mathbb{R}^{\eta_i \times 2}$ is a set of unknown NN weights, and ϵ_i is an approximation error.

The estimation, $\hat{f}_i$, is given by

$$\hat{f}_i = \hat{W}_i^T \varphi_i(z_i) \quad (13)$$

where $\hat{W}$ is the tunable NN weights matrix, the activation function $\varphi_i(z)$ is selected as

$$\varphi_i(z_i) = \exp \left[\frac{-(z_i - \mu_i)^T (z_i - \mu_i)}{\Phi_i^2} \right] \quad (14)$$

and $\mu_i = [\mu_{i1}, \dots, \mu_{i4}]^T$ is the center of the activation function and Φ_i is the width of the Gaussian function.

ASSUMPTION 6 Unknown, ideal NN weights matrix is bounded such that $\|W_i\|_F \leq W_M$, with a fixed bound W_M .

ASSUMPTION 7 The NN approximation error $\epsilon(z)$ over a compact set Ω is bounded by $\|\epsilon_i(z)\| \leq \epsilon_M$.

REMARK 2 Consider the vector function $f_i(\tilde{\mathbf{x}})$ in (6) to be smooth and defined on a compact set of Ω_i . Since f_i is a

smooth function and Weierstrass high-order approximation theorem can be applied, there exists a sufficient number of neurons such that NN approximation error ϵ_i is bounded by a fixed bound $\|\epsilon_i\| \leq \epsilon_M$ [40], [48].

In the following, we use the preliminary result of [29] and extend it to study the formation control and target tracking of multiagent systems modeled by second-order nonlinear dynamics.

III. MAIN RESULT

A. Control Algorithm Design

Consider a multiagent system (6) that is modeled by an undirected graph. The distance error is given by

$$e_{ij} = \|\tilde{p}_{ij}\| - d_{ij} \quad (15)$$

where $\tilde{p}_{ij} = p_i - p_j$. From (6), distance error dynamics is

$$\dot{e}_{ij} = \frac{\tilde{p}_{ij}^T(\dot{p}_i - \dot{p}_j)}{\|\tilde{p}_{ij}\|} = \frac{\tilde{p}_{ij}^T(v_i - v_j)}{e_{ij} + d_{ij}}. \quad (16)$$

Define $\mathbf{e} = [\dots, e_{ij}, \dots]^T \in \mathbb{R}^{|E|}$, auxiliary variable q_{ij} as

$$q_{ij} = e_{ij}(e_{ij} + 2d_{ij}) \quad (17)$$

and the energy function as $M_1(\mathbf{e}) = \sum_{(i,j) \in E} M_{ij}(e_{ij})$ [1], where

$$M_{ij}(e_{ij}) = \frac{1}{4}q_{ij}^2 = \frac{1}{4}e_{ij}^2(e_{ij} + 2d_{ij})^2. \quad (18)$$

Considering equations (16) and (18), $\dot{M}_1(\mathbf{e})$ is given by

$$\dot{M}_1(\mathbf{e}) = \sum \frac{\partial M_{ij}}{\partial e_{ij}} \frac{\tilde{p}_{ij}^T(v_i - v_j)}{e_{ij} + d_{ij}}. \quad (19)$$

Let us define $\beta(\mathbf{e})$ as

$$\beta(\mathbf{e}) = \left(\dots, \frac{\partial M_{ij}}{\partial e_{ij}}, \dots \right) = (\dots, e_{ij}(e_{ij} + 2d_{ij}), \dots) \quad (20)$$

where pair $(i, j) \in E$ and $\beta(\mathbf{e}) \in \mathbb{R}^{|E|}$. As shown in [45], it follows from (3) and (16) that

$$\dot{M}_1 = \beta^T(\mathbf{e})R_p\mathbf{x}_2. \quad (21)$$

Using the backstepping technique [1], we define $\mathbf{s} = [s_1^T, \dots, s_N^T]^T \in \mathbb{R}^{2N}$ as

$$\mathbf{s} = \mathbf{x}_2 - \tau. \quad (22)$$

The auxiliary variable τ is given by

$$\tau = u_s + \mathbf{1}_N \otimes (v_r + k_r e_r) \quad (23)$$

where $\tau = [\tau_1^T, \dots, \tau_N^T]^T \in \mathbb{R}^{2N}$ and $u_s = -k_v R_p^T \beta(\mathbf{e}) \in \mathbb{R}^{2N}$, k_r and k_v are positive constants. One can express u_s elementwise as $u_s = [u_{s_1}^T, \dots, u_{s_N}^T]^T \in \mathbb{R}^{2N}$. The first term in (23) is responsible for achieving the desired formation of agents, while the second term is used for tracking the target by the leader and other agents. Define the potential function

$$M_2 = M_1(\mathbf{e}) + \frac{1}{2}\mathbf{s}^T \mathbf{s}. \quad (24)$$

Taking a time-derivative of (24), using Lemma 1, and (7), (23), yields

$$\begin{aligned} \dot{M}_2 &= \beta^T(\mathbf{e})R_p\mathbf{x}_2 + \mathbf{s}^T \dot{\mathbf{s}} \\ &= \beta^T(\mathbf{e})R_p\tau + \mathbf{s}^T [f(\mathbf{x}) + g(\mathbf{x})u + w \\ &\quad + R_p^T \beta(\mathbf{e}) - \dot{\tau}] \\ &= -k_v \beta^T(\mathbf{e})R_p R_p^T \beta(\mathbf{e}) + \mathbf{s}^T [f(\mathbf{x}) + g(\mathbf{x})u \\ &\quad + w + R_p^T \beta(\mathbf{e}) - \dot{\tau}]. \end{aligned} \quad (25)$$

Using Cauchy–Schwarz inequality, one can write

$$\begin{aligned} \mathbf{s}^T f(\mathbf{x}) &\leq \sum_{i=1}^N s_i^T f_i(\tilde{\mathbf{x}}_i) \leq \sum_{i=1}^N s_i^T (W_i^T \varphi_i(\tilde{\mathbf{x}}_i) + \epsilon_i) \\ &\leq \sum_{i=1}^N (\|s_i\| \|W_i\| \|\varphi_i(\tilde{\mathbf{x}}_i)\| + \|s_i\| \|\epsilon_i\|). \end{aligned} \quad (26)$$

Using the fact that $\|\cdot\|_F \geq \|\cdot\|$, from (12) and Young's inequality, one can write (26) as

$$\begin{aligned} \mathbf{s}^T f(\mathbf{x}) &\leq \sum_{i=1}^N \left(\|W_i\|_F^2 (\|s_i\| \|\varphi_i(\tilde{\mathbf{x}}_i)\|)^2 + \frac{1}{4\alpha_i^2} \right. \\ &\quad \left. + \|s_i\|^2 + \frac{1}{4\alpha_i^2} \|\epsilon_i\|^2 \right) \end{aligned} \quad (27)$$

where $\alpha_i \forall i \in \{1, \dots, N\}$ is a positive constant. Let us define $\varpi = [\varpi_1, \dots, \varpi_N]^T$, where the norm of the NN weights matrix is given as $\varpi_i = \|W_i\|_F^2$ for $i \in \{1, \dots, N\}$, and $\hat{\varpi}_i$ is an estimation of ϖ_i [26]. From Assumption 6, it is clear that ϖ_i is bounded by W_M . Define the NN weights' norm estimation error as $\tilde{\varpi} = \varpi - \hat{\varpi}$, where $\hat{\varpi} = [\hat{\varpi}_1, \dots, \hat{\varpi}_N]^T \in \mathbb{R}^N$. Define $\Xi = R_p^T \beta(\mathbf{e}) \in \mathbb{R}^{2N}$, where vector Ξ can be written as $\Xi = [\Xi_1^T, \dots, \Xi_N^T]^T$, with $\Xi_i = \sum_{j \in N_i} p_{ij} q_{ij} \in \mathbb{R}^2$. From Assumption 2 and following the same procedure as (27), one can obtain the following inequalities:

$$\begin{aligned} \mathbf{s}^T w &\leq \sum_{i=1}^N \left((\|s_i\|)^2 + \frac{1}{4\eta_i^2} w_M^2 \right) \\ \mathbf{s}^T R_p^T \beta(\mathbf{e}) &\leq \sum_{i=1}^N \left((\|s_i\| \|\Xi_i\|)^2 + \frac{1}{4\Upsilon_i^2} \right) \\ \mathbf{s}^T \dot{\tau} &\leq \sum_{i=1}^N \left((\|s_i\| \|\dot{u}_{s_i} + k_r \dot{e}_r\|)^2 + \frac{1}{4\psi_i^2} \right) + s_i^T \dot{v}_r \end{aligned} \quad (28)$$

where $\Upsilon_i, \psi_i, \eta_i, \forall i \in \{1, \dots, N\}$ are the positive constants. Substituting (27) and (28) into (25), one can obtain

$$\begin{aligned} \dot{M}_2 &= -k_v \beta^T(\mathbf{e})R_p R_p^T \beta(\mathbf{e}) + \mathbf{s}^T [f(\mathbf{x}) + g(\mathbf{x})u + w \\ &\quad + R_p^T \beta(\mathbf{e}) - \dot{\tau}] \\ &\leq -k_v \beta^T(\mathbf{e})R_p R_p^T \beta(\mathbf{e}) + \sum_{i=1}^N [\|s_i\|^2 (\varpi_i \|\varphi_i(\tilde{\mathbf{x}}_i)\|)^2 \\ &\quad + \|\Xi_i\|^2 + 2 + (\|\dot{u}_{s_i} + k_r \dot{e}_r\|)^2] + s_i^T \dot{v}_r \end{aligned}$$

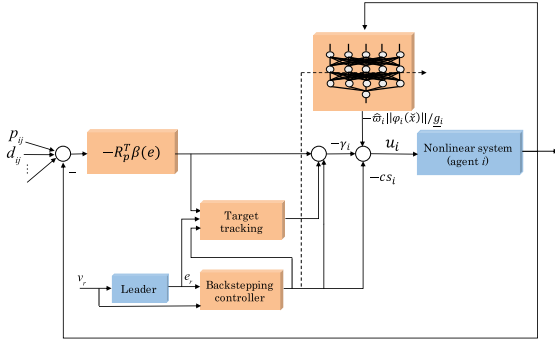


Fig. 4. Block diagram of the proposed control scheme for each agent.

$$+ s_i^T g_i(\tilde{\mathbf{x}}_i) u_i + \sum_{i=1}^N \left(\frac{1}{4\varrho_i^2} + \frac{1}{4\alpha_i^2} \|\epsilon_i\|^2 + \frac{1}{4\eta_i^2} w_M^2 + \frac{1}{4\Upsilon_i^2} + \frac{1}{4\psi_i^2} \right). \quad (29)$$

Consider the following energy function:

$$V = M_2 + \frac{1}{2} \tilde{\omega}^T F^{-1} \tilde{\omega} \quad (30)$$

where the matrix $F = \Pi_i I_N \in \mathbb{R}^{N \times N}$ is a positive-definite matrix, i.e., $\Pi_i \in \mathbb{R}^+$, $\forall i \in \{1, \dots, N\}$. We propose the control law as follows:

$$u_i = -cs_i - \gamma_i - \frac{1}{\underline{g}_i} \hat{\omega}_i \|\varphi_i(\tilde{\mathbf{x}}_i)\|^2 s_i \quad (31)$$

where constant c is a positive scalar. Moreover, the term γ_i is given as

$$\gamma_i = \frac{1}{\underline{g}_i} (\|\dot{u}_{s_i} + k_r \dot{e}_r\|^2 + \|\Xi_i\|^2) s_i + \frac{b}{\underline{g}_i} \text{sgn}(s_i) \quad (32)$$

with b being a positive constant. The tuning law of $\hat{\omega}_i$ is given by

$$\dot{\hat{\omega}}_i = \Pi_i \|\varphi_i(\tilde{\mathbf{x}}_i)\|^2 \|s_i\|^2 - \kappa_i \Pi_i \hat{\omega}_i. \quad (33)$$

The control input signal is comprised of tracking error between the leader and the target, a backstepping controller, and an NN-based component to compensate for the unknown nonlinearity of the system dynamics, (see Fig. 4). The controller is developed based on relative positions and velocities of neighboring agents only; therefore, the proposed controller is distributed [31]. The control input signal (31) for each agent can be written as follows:

$$u_i = -cs_i - \frac{1}{\underline{g}_i} \hat{\omega}_i \|\varphi_i(\tilde{\mathbf{x}}_i)\|^2 s_i - \left\| \sum_{j \in N_i} [k_v(q_{ij} I_2 + 2\tilde{p}_{ij} \tilde{p}_{ij}^T) v_{ij}] - k_r \dot{e}_r \right\|^2 s_i + \frac{1}{\underline{g}_i} \left\| \sum_{j \in N_i} \tilde{p}_{ij} q_{ij} \right\|^2 s_i + \frac{b}{\underline{g}_i} \text{sgn}(s_i) \quad (34)$$

with

$$s_i = v_i - v_r - k_r e_r + k_v \sum_{j \in N_i} \tilde{p}_{ij} q_{ij}. \quad (35)$$

REMARK 3 The control input signal (34) includes the relative position and relative velocity vectors between agent i and its neighboring agents j . This over-sensing is necessary for the distance-based formation [31], [49]–[52] and the volume of the over-sensing can be reduced by considering a minimally rigid framework. The relative position between the agent i and its arbitrary neighboring agent j can be presented by the distance and angle sensing between agents. This angle between agent i and j can be obtained through information such as heading of neighboring agent for a rigid framework [53]. The agent is also capable to calculate the relative velocity vector by differentiating the relative position.

B. Stability Analysis

In this section, we show that the proposed control law (31) can be used for nonlinear multiagent systems (7) with guaranteed stability in terms of the agents formation.

THEOREM 2 Consider the framework $F = (G, p)$ to be modeled by an undirected, infinitesimally, and minimally rigid graph. Select the control input signals as (31) and NN weights' norm tuning law as (33). Under Assumptions 1–7, choose $b \geq \sqrt{2} \|\dot{v}_r\|_\infty$ and $c > \frac{2}{g}$. Then, for initial conditions that belong to the compact set Ω_0 defined by

$$\Omega_0 = \{\mathbf{x}_1(0), p_r(0) | p_r(0) \in \Omega_T, d(\mathbf{x}_1, \text{Iso}(F^*)) < d(\mathbf{x}_1, \text{Amb}(F^*))\} \sum_{(i,j) \in E^*} (\|p_{ij}\|^2 - d_{ij}^2)^2 \leq \delta \quad (36)$$

the distance errors and the NN weights' norm estimation errors are UUB. The leader follows the target and followers maintain the desired formation (F^*) with the leader.

PROOF Note that the closed-loop system with (31)–(33) has a discontinuity due to $\text{sgn}(s_i)$ in (32) at $s_i = 0$. We choose the Lyapunov function candidate as (30). Let $\zeta_i = \mathcal{F}_i(\zeta_i, t)$ be the closed-loop system with respect to multiagent systems with (31)–(33). Following the procedure in [30], [31], the time-derivative of Lyapunov function (30) is given by

$$\dot{V} \stackrel{a.e.}{\in} \sum_{i=1}^N \frac{\partial V}{\partial \zeta_i} K_i[\mathcal{F}_i](\zeta_i, t) \subset \dot{M}_2 + \tilde{\omega}^T F^{-1} \dot{\tilde{\omega}}. \quad (37)$$

Recall the definition of $\tilde{\omega}$ and consider the Assumption 6. By considering (29) and substituting the proposed control input (31), using the NN tuning law (33), one obtains

$$\dot{V} \leq -k_v \beta^T(e) R_p R_p^T \beta(e) + \sum_{i=1}^N [\|s_i\|^2 (\tilde{\omega}_i \|\varphi_i(\tilde{\mathbf{x}}_i)\|^2 + 2 + \|\Xi_i\|^2 + \|\dot{u}_{s_i} + k_r \dot{e}_r\|^2) + s_i^T \dot{v}_r] + \sum_{i=1}^N \left(\frac{1}{4\varrho_i^2} + \frac{1}{4\alpha_i^2} \|\epsilon_i\|^2 + \frac{1}{4\eta_i^2} w_M^2 + \frac{1}{4\Upsilon_i^2} + \frac{1}{4\psi_i^2} \right) + \sum_{i=1}^N \tilde{\omega}_i (-\|\varphi_i(\tilde{\mathbf{x}}_i)\|^2 \|s_i\|^2 + \kappa_i \hat{\omega}_i)$$

$$+ \sum_{i=1}^N s_i^T g_i(\tilde{\mathbf{x}}_i) \left(-cs_i - \frac{b}{\underline{g}_i} \text{sgn}(s_i) - \frac{1}{\underline{g}_i} (\|\dot{u}_{s_i} + k_r e_r\|^2 + \|\Xi_i\|^2 + \hat{\omega}_i \|\varphi_i(\tilde{\mathbf{x}}_i)\|^2) s_i \right). \quad (38)$$

From Assumption 1, matrix $g_i(\tilde{\mathbf{x}}_i)$ is a full-rank and positive-definite matrix. Using the fact that $g_i(\tilde{\mathbf{x}}_i)g_i^{-1}(\tilde{\mathbf{x}}_i) = I_2$, one can have $\frac{\|g_i(\tilde{\mathbf{x}}_i)\|}{\underline{g}_i} \geq 1$. Considering the fact that $s_i^T \text{sgn}(s_i) \geq 0$, the following inequality holds:

$$- \frac{b}{\underline{g}_i} s_i^T g_i(\tilde{\mathbf{x}}_i) \text{sgn}(s_i) \leq -bs_i^T \text{sgn}(s_i). \quad (39)$$

From (39), and by simplifying and rearranging (38), one has

$$\begin{aligned} \dot{V} \leq & -k_v \beta^T(\mathbf{e}) R_p R_p^T \beta(\mathbf{e}) - \sum_{i=1}^N ((c\underline{g}_i - 2)\|s_i\|^2 \\ & + \|s_i\|(\sqrt{2}\|\dot{v}_r\|_\infty - b)) + \sum_{i=1}^N \left(\frac{1}{4Q_i^2} + \frac{1}{4\alpha_i^2} \|\epsilon_i\|^2 \right. \\ & \left. + \frac{1}{4\eta_i^2} w_M^2 + \frac{1}{4\Upsilon_i^2} + \frac{1}{4\psi_i^2} \right) + \sum_{i=1}^N \kappa_i \tilde{\omega}_i(\hat{\omega}_i). \end{aligned} \quad (40)$$

From (28) and (40), one can notice that the selection of b is important to counteract effect of the disturbance in the system dynamics. Let us define $\underline{\kappa} = \min(\kappa_1, \dots, \kappa_N)$. Applying Young's inequality to the last term, yields

$$\tilde{\omega}_i \hat{\omega}_i \leq -\frac{\tilde{\omega}_i^2}{2} + \frac{\omega_i^2}{2}. \quad (41)$$

Recalling the definition of $\underline{g}$, using $b \geq \sqrt{2}\|\dot{v}_r\|_\infty$, and Assumptions 6–7, one has

$$\dot{V} \leq -k_v \beta^T(\mathbf{e}) R_p R_p^T \beta(\mathbf{e}) - (c\underline{g} - 2)\|s\|^2 - \frac{\kappa}{2} \|\tilde{\omega}\|^2 + q \quad (42)$$

where q is written as

$$q = \sum_{i=1}^N \left(\frac{1}{4Q_i^2} + \frac{1}{4\alpha_i^2} \epsilon_M^2 + \frac{1}{4\eta_i^2} w_M^2 + \frac{1}{4\Upsilon_i^2} + \frac{1}{4\psi_i^2} + \kappa_i \frac{W_M^4}{2} \right). \quad (43)$$

Based on Assumptions 1–7, the time-derivative of Lyapunov function (38) is bounded by

$$\dot{V} \leq -k_v \underline{\sigma}(R_p) \|\beta(\mathbf{e})\|^2 - (c\underline{g} - 2)\|s\|^2 - \frac{\kappa}{2} \|\tilde{\omega}\|^2 + q \quad (44)$$

or equivalently

$$\dot{V} \leq - \left[\|\beta(\mathbf{e})\| \quad \|s\| \quad \|\tilde{\omega}\| \right] \underbrace{\begin{bmatrix} k_v \underline{\sigma}(R_p) & 0 & 0 \\ 0 & c\underline{g} - 2 & 0 \\ 0 & 0 & \frac{\kappa}{2} \end{bmatrix}}_Q \begin{bmatrix} \|\beta(\mathbf{e})\| \\ \|s\| \\ \|\tilde{\omega}\| \end{bmatrix} + q \quad (45)$$

where $\underline{\sigma}(\cdot)$ denotes the minimum singular values of a matrix. Defining $z = [\|\beta(\mathbf{e})\| \quad \|s\| \quad \|\tilde{\omega}\|]^T$, (45) can be rewritten as

$$\dot{V} \leq -z^T Q z + q. \quad (46)$$

For $\dot{V} \leq 0$ one requires $Q \geq 0$ and

$$\|z\|^2 > \frac{q}{\underline{\sigma}(Q)}. \quad (47)$$

From (30), one can write the following inequality for V :

$$\begin{aligned} \frac{1}{2} [\|\beta(\mathbf{e})\| \quad \|s\| \quad \|\tilde{\omega}\|] & \overbrace{\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{\Pi_{\max}} \end{bmatrix}}^{T_1} \begin{bmatrix} \|\beta(\mathbf{e})\| \\ \|s\| \\ \|\tilde{\omega}\| \end{bmatrix} \\ & \leq V \leq \\ \frac{1}{2} [\|\beta(\mathbf{e})\| \quad \|s\| \quad \|\tilde{\omega}\|] & \underbrace{\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{\Pi_{\min}} \end{bmatrix}}_{T_2} \begin{bmatrix} \|\beta(\mathbf{e})\| \\ \|s\| \\ \|\tilde{\omega}\| \end{bmatrix} \end{aligned} \quad (48)$$

where $\underline{\sigma}(T_1) = \min(\frac{1}{2}, \frac{1}{\Pi_{\max}})$ and $\bar{\sigma}(T_2) = \max(1, \frac{1}{\Pi_{\min}})$, with $\bar{\sigma}(\cdot)$ denoting the maximum singular values of a matrix. Therefore, one has

$$\frac{1}{2} \underline{\sigma}(T_1) \|z\|^2 \leq V \leq \frac{1}{2} \bar{\sigma}(T_2) \|z\|^2. \quad (49)$$

From (49), we conclude that

$$V \geq \frac{\underline{\sigma}(T_1)}{2\underline{\sigma}(Q)} q \quad (50)$$

implies (47), and therefore, $z(t)$ is UUB [54]. As for any vector z , one has $\|z\|_1 \geq \|z\|_2 \geq \dots \geq \|z\|_\infty$, and sufficient conditions for inequity (47) are [24]

$$\|\beta(\mathbf{e})\| > \sqrt{\frac{q}{\underline{\sigma}(Q)}} \quad (51)$$

or

$$\|s\| > \sqrt{\frac{q}{\underline{\sigma}(Q)}} \quad (52)$$

or

$$\|\tilde{\omega}\| > \sqrt{\frac{q}{\underline{\sigma}(Q)}}. \quad (53)$$

This shows that $\|\beta(\mathbf{e})\|$, $\|s\|$, $\|\tilde{\omega}\|$ are UUB. Using (46) and (49), one can write

$$\dot{V} \leq -a_1 V + a_2 \quad (54)$$

with $a_1 \triangleq 2\underline{\sigma}(Q)/\bar{\sigma}(T_2)$ and $a_2 \triangleq q$. From (54), it follows that:

$$\begin{aligned} V(t) & \leq V(0)e^{-a_1 t} + \frac{a_2}{a_1} (1 - e^{-a_1 t}) \\ & \leq V(0) + \frac{a_2}{a_1}. \end{aligned} \quad (55)$$

Using inequality (49) yields

$$\|\beta(\mathbf{e})\| \leq \|z(t)\|$$

$$\leq \sqrt{\frac{\bar{\sigma}(T_2)}{\underline{\sigma}(T_1)} (\|\beta(e(0))\|^2 + \|\mathbf{s}(0)\|^2 + \|\tilde{\omega}(0)\|^2)} + \sqrt{\frac{\bar{\sigma}(T_2)q}{\underline{\sigma}(T_1)\underline{\sigma}(Q)}} \triangleq \rho \quad (56)$$

and

$$\|\mathbf{s}\| \leq \|z(t)\| \leq \rho \quad (57)$$

and

$$\|\hat{\omega}\| \leq \|z(t)\| \leq \rho. \quad (58)$$

From (56), we have

$$\|\beta(\mathbf{e})\| = \sqrt{\sum_{(i,j) \in E} (e_{ij}^2 + 2d_{ij}e_{ij})^2} \leq \rho. \quad (59)$$

By substituting e_{ij} in inequality (59)

$$\|\beta(\mathbf{e})\| = \sqrt{\sum_{(i,j) \in E} ((\|\tilde{p}_{ij}\| - d_{ij})^2 (\|\tilde{p}_{ij}\| + d_{ij})^2)} \leq \rho. \quad (60)$$

From (60), for the edge $(i, j) \in E$ one can write

$$\|\tilde{p}_{ij}\|^2 - d_{ij}^2 \leq \rho. \quad (61)$$

Using Assumption 3, the upper bound is given by

$$\|\tilde{p}_{ij}\| \leq \sqrt{\rho + D^2}. \quad (62)$$

As desired distances d_{ij} 's are considered bounded, the inequality (62) implies boundedness of the distance error dynamics (e_{ij}) for all $(i, j) \in E$. Thus, interagent distance error as well as NN weights' norm estimation error remains bounded. ■

LEMMA 3 Given the bounded initial conditions for $p_r(0)$ and $p_1(0)$, the tracking error between the leader and the target is bounded by $\|e_r(t)\| \leq \bar{\xi}$, with the bound

$$\bar{\xi} = \|e_r(0)\| + \frac{\sqrt{2}\rho}{k_r}(1 + k_v\bar{\sigma}(R_p)). \quad (63)$$

PROOF From (22) and (23), the velocity of the leader (first agent) can be written as

$$v_1 = \tau_1 + s_1 = u_{s1} + v_r + k_r e_r + s_1. \quad (64)$$

The target tracking error dynamics is

$$\dot{e}_r = v_r - v_1 = -k_r e_r - (s_1 - u_{s1}). \quad (65)$$

From (57) and (59), it follows that

$$\dot{e}_r \leq -k_r e_r + \rho(1 + k_v\bar{\sigma}(R_p)) \otimes \mathbf{1}_2. \quad (66)$$

Multiplying both sides by $e^{k_r t}$ yields

$$\frac{d}{dt}(e_r e^{k_r t}) \leq \rho(1 + k_v\bar{\sigma}(R_p))e^{k_r t} \otimes \mathbf{1}_2. \quad (67)$$

By taking integral over $[0, t]$, we have

$$e_r(t) \leq \left[e_r(0) - \left(\frac{\rho(1 + k_v\bar{\sigma}(R_p))}{k_r} \right) \right] e^{-k_r t} \otimes \mathbf{1}_2 + \left(\frac{\rho(1 + k_v\bar{\sigma}(R_p))}{k_r} \right) \otimes \mathbf{1}_2. \quad (68)$$

We can conclude that

$$e_r(t) \leq e_r(0) + \left(\frac{\rho(1 + k_v\bar{\sigma}(R_p))}{k_r} \right) \otimes \mathbf{1}_2. \quad (69)$$

Equivalently, $\|e_r(t)\| \leq \bar{\xi}$. ■

We provide a bound over the states of second-order nonlinear multiagent systems defined by (6). From (10), it follows that:

$$\|p_1\| = \|p_r - e_r\| \leq \|p_r\| + \|e_r\| \quad (70)$$

which yields

$$\|p_1\| \leq \bar{P}_r + \bar{\xi}. \quad (71)$$

As the framework is considered infinitesimally and minimally rigid, there exists at least one neighbor of the leader, say agent j . Therefore, from triangle inequality and (62), we have

$$\exists j \in N_1 \mid \|\tilde{p}_{j1}\| = \|p_j - p_1\| \geq \|p_j\| - \|p_1\|. \quad (72)$$

From (62) and (72), we can write

$$\begin{aligned} \|p_j\| - \|p_1\| &\leq \sqrt{\rho + D^2} \\ \|p_j\| &\leq \|p_1\| + \sqrt{\rho + D^2} \\ \|p_j\| &\leq \bar{P}_r + \bar{\xi} + \sqrt{\rho + D^2}. \end{aligned} \quad (73)$$

Since the framework is rigid, there exists another vertex $k \in N_j$. Using (62) and (73), we have

$$\|p_k\| \leq \bar{P}_r + \bar{\xi} + 2\sqrt{\rho + D^2}. \quad (74)$$

Thus, for all positions there is a bound ($\forall i \in V$, $\exists \bar{P}_i \mid \|p_i\| \leq \bar{P}_i$); equivalently, $\|\mathbf{x}_1\| \leq \|\bar{P}\|$, where $\bar{P} = [\bar{P}_1, \dots, \bar{P}_N]^T$. From (22), and using triangle inequality, we have

$$\begin{aligned} \|\mathbf{x}_2\| &\leq \|\mathbf{s}\| + \|\tau\| \\ \|\mathbf{x}_2\| &\leq \|\mathbf{s}\| + \|\mathbf{u}_s + \mathbf{1}_N \otimes (v_r + k_r e_r)\|. \end{aligned} \quad (75)$$

Using inequalities (49), (57), and $\|e_r(t)\| \leq \bar{\xi}$, we conclude

$$\|\mathbf{x}_2\| \leq \rho(1 + k_v\bar{\sigma}(R_p)) + \sqrt{N}(\bar{V}_r + k_r\bar{\xi}) \triangleq \bar{V}. \quad (76)$$

Since the velocity of the target is considered to be bounded (Assumption 4), for $t \geq 0$ there exists a compact set of Ω , where $\Omega = \{\mathbf{x} \mid \|\mathbf{x}_1\| \leq \bar{P}, \|\mathbf{x}_2\| \leq \bar{V}\}$.

COROLLARY 1 Let the required framework be modeled by an undirected, infinitesimally, and minimally rigid graph where each agent is described by a second-order nonlinear system in Brunovsky form as follows:

$$\begin{aligned} \dot{p}_i &= v_i \\ \dot{v}_i &= f_i(p_i, v_i) + u_i(t) + w_i(t) \\ i &= 1, 2, \dots, N. \end{aligned} \quad (77)$$

Select the control input signals as (31) and NN weights' norm tuning law as (33), under Assumptions 2-7, choose $b \geq \sqrt{2}\|\dot{v}_r\|_\infty$, $c > 2$, and $\underline{g}_i = 1$. Then, for initial conditions that belong to the compact set Ω_0 , the NN weights'

norm estimation error and interagent distance errors are UUB. The leader tracks the target, and other agents maintaining the desired formation follow the leader.

PROOF Set $g_i = 1$, then it is not difficult to see the proof of Theorem 2 still holds for a class of nonlinear multiagent systems given as (77).

REMARK 4 Inequalities (56) and (58) provide bounds for interagent distance errors and the NN weights' norm estimation errors. As the leader (first agent) tracks the moving target within a bounded trajectory, the leader's position remains bounded by (71). The control law guarantees the boundedness of interagent distance errors, and other agents remain in their desired formation. It has been shown that the velocities of agents remain bounded as the velocity of the target is bounded. Using the proposed control, and based on Assumptions 3–5, all positions and velocities of the agents remain bounded. We use this bounded region to form a compact set Ω , where NN approximation property holds.

REMARK 5 Signal γ_i in (32) has a discontinuous right-hand side part. Therefore, the existence of a general solution should be checked through differential inclusions. As the sign function is a bounded function, the Filippov solution exists, [31], [55]. Consequently, the time-derivative of Lyapunov function (30) belongs to a closed, upper semicontinuous set-valued map [56]. Thus, the Lyapunov function is continuously differentiable, and the proof is valid. Moreover, to alleviate the undesired chattering effect caused by the sign function, a hyperbolic tangent can be used in practical applications.

REMARK 6 The proposed control law guarantees that the multiagent system (6) converges to the equilibrium within the set Ω_0 . Thus, the distance errors remain in the invariant set Ω_0 [31] and $\tilde{p}_{ij} \neq 0$. Moreover, the multiagent system (6) converges to the equilibrium inside this invariant set, thus preventing the flip ambiguity.

IV. SIMULATION RESULTS

We conducted multiple simulations in order to evaluate the performance of the proposed method for distance-based formation control applied to second-order nonlinear multiagent systems. In each example, the results of the proposed distance-based method are compared with the results of existing displacement-based methods. Also, to evaluate and quantify the performance of each method, a performance index is introduced.

EXAMPLE 1 Consider a nonlinear multiagent system with $N = 5$ agents in a plane where the dynamics of each agent is given by

$$\begin{aligned} \dot{p}_i(t) &= v_i(t) \\ \dot{v}_i(t) &= \begin{pmatrix} c_{i1} v_{i1}(t) \sin(d_{i1} p_{i1}(t)) \\ c_{i2} v_{i2}(t) \cos(d_{i2} p_{i2}(t)) \end{pmatrix} + u_i(t) + w_i(t) \\ i &= 1, 2, \dots, 5. \end{aligned} \quad (78)$$

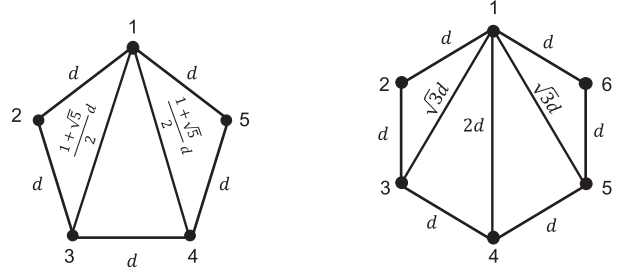


Fig. 5. Communication topology and desired distances of multi-agent systems: infinitesimally and minimally rigid frameworks of five agents (left) and six agents (right) in 2-D.

TABLE I
Values of c_{i1} , c_{i2} and d_{i1} , d_{i2} in the i th Agent

i	1	2	3	4	5
c_{i1}	0.2	0.4	-0.6	0.7	-0.8
c_{i2}	-0.3	0.6	0.5	0.5	0.6
d_{i1}	0.4	0.7	0.3	-0.4	-0.7
d_{i2}	0.3	-0.4	0.4	0.8	-0.5

The goal is to form the desired formation, shown in Fig. 5 (left), while the agents follow the leader which is tracking a target within a bounded trajectory. To this end, the proposed control law in Corollary 1 is applied.

Vectors $p_i(t) = [p_{i1}, p_{i2}]^T$ and $v_i(t) = [v_{i1}, v_{i2}]^T$ represent the location and velocity of i th agent in x and y directions, respectively. Parameters c_{i1} , c_{i2} , d_{i1} , d_{i2} of each agent are given in Table I. The initial conditions for these five agents are: $p_1(0) = (0, 2.1)^T$, $v_1(0) = (0.5, 0.5)^T$, $p_2(0) = (-1, 1.2)^T$, $v_2(0) = (-1, 0.8)^T$, $p_3(0) = (-2, -1.5)^T$, $v_3(0) = (0.3, 0.5)^T$, $p_4(0) = (0.2, -1.2)^T$, $v_4(0) = (0.5, 1)^T$, $p_5(0) = (0.5, 1)^T$, $v_5(0) = (-1, -1)^T$. The disturbance $w_i = 0.1 \cos(t) \delta(t)$, where $\delta(t) \in \mathbb{R}^2$ is a random variable with the normal distribution $\mathcal{N}(0, 1)$ borrowed from [57], [58]. The target velocity and initial position are $v_r = [0.5, 0.5 \cos(t/2)]^T$ and $p_r(0) = [0, 0]^T$, respectively.

We selected constants $b = 3$ and $k_r = 2$. RBFNN is selected with nine neurons, centers μ_j 's are evenly spaced for each agent, and the values of constants are: $\kappa = 0.5$, $F_i = I_9$, $c = 5$, and $k_v = 0.5$. The desired distances and communication topology are given in Fig. 5 (left) with $d = 1$. Figs. 6–10 show results of the proposed control law. Fig. 6 shows how agents form an equilateral pentagon, where each side is equal to the desired distance. Fig. 7 represents the trajectories of each agent and the target in x and y directions. Velocity of each agent and the target in x and y directions are shown in 8.

In order to compare the results with a displacement-based method, the modified version of [24] is simulated. We compared distance (edge) errors in Fig. 9 and the target tracking errors of the leader along x and y directions and its time-derivative in Fig. 10. We use an average absolute

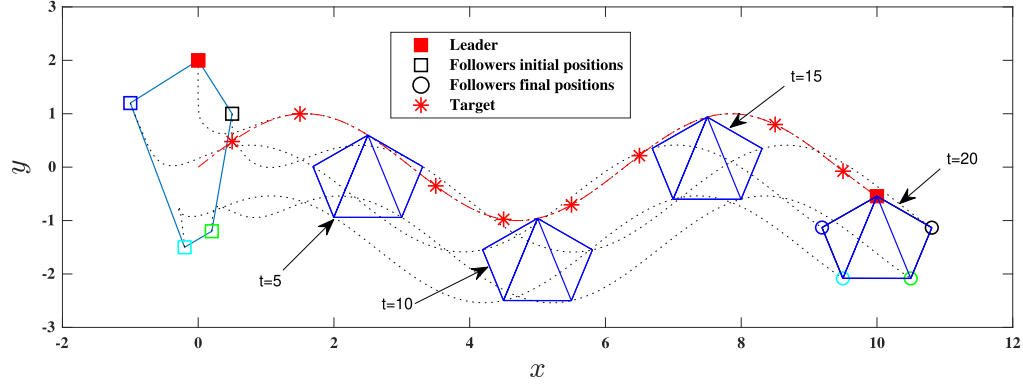


Fig. 6. Desired formation of equilateral pentagon with side length $d = 1$.

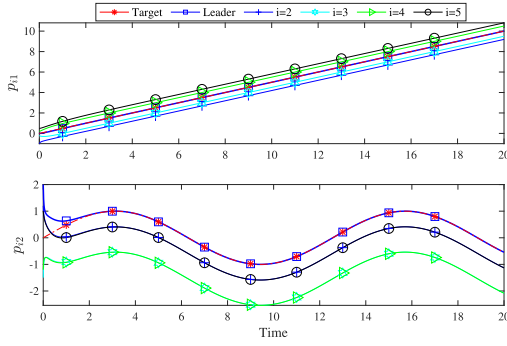


Fig. 7. Trajectories of agents and the target along x and y directions.

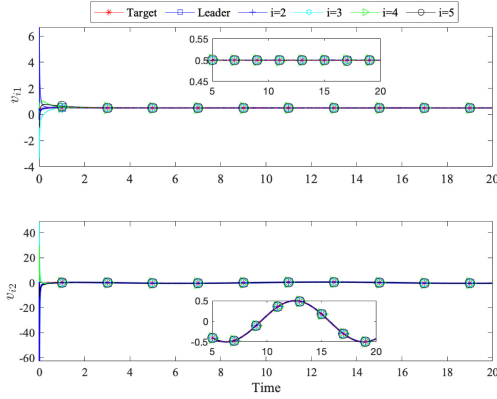


Fig. 8. Speeds of agents and the target along x and y directions.

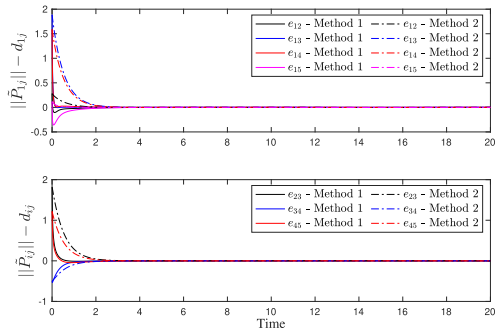


Fig. 9. Distance errors e_{ij} , $(i, j) \in E$ of the proposed method (Method 1) and the modified displacement-based method (Method 2) [24].

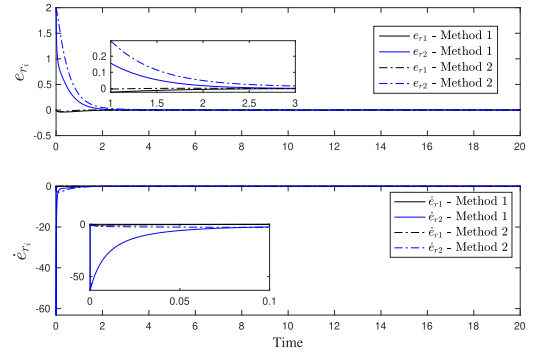


Fig. 10. Target tracking error of the leader and its time-derivative for the proposed method and the modified displacement-based method [24] along x and y directions.

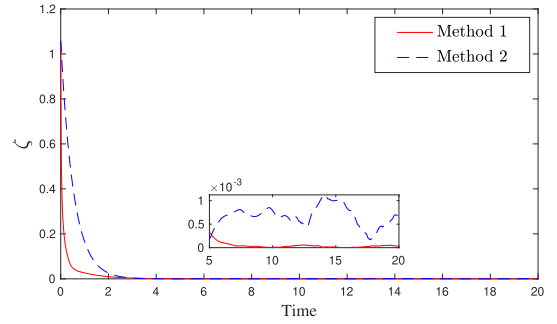


Fig. 11. Performance comparison (AFDE) between the proposed distance-based method and the modified displacement-based method [24].

formation distance error (AFDE) $\zeta(t)$ function [30]

$$\zeta(t) = \frac{1}{|E|} \sum_{(i,j) \in E} |(|\tilde{p}_{ij}| - d_{ij})| \quad (79)$$

which is plotted in Fig. 11. The proposed method has an improved AFDE compared to the displacement method.

EXAMPLE 2 Consider a multiagent system consisting of $N = 6$ two-link robot arms. The goal is to form the desired formation, shown in Fig. 5 (right), while the agents follow the leader that tracks a target within a bounded trajectory.

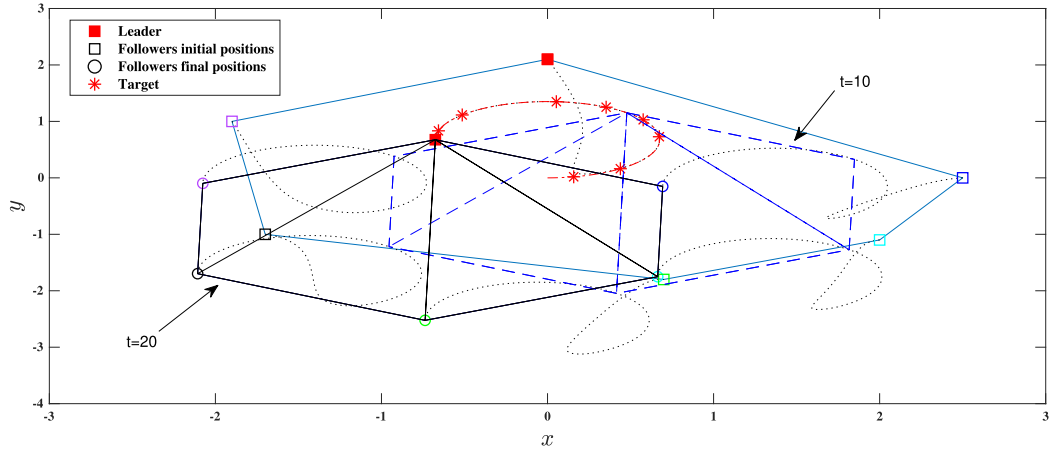


Fig. 12. Desired formation of equilateral hexagon in 2-D with side length $d = 1.6$.

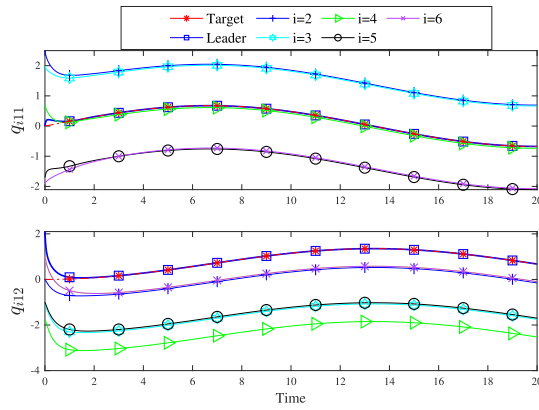


Fig. 13. Trajectories of the agents and the target along x and y directions.

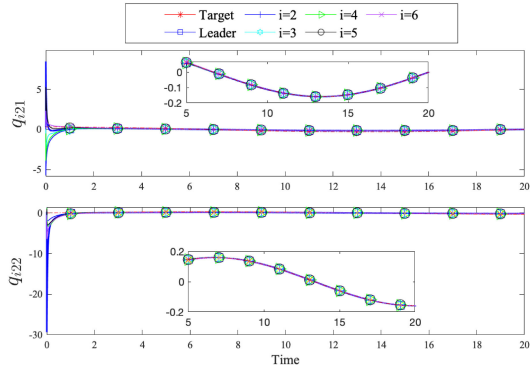


Fig. 14. Velocities of the agents and the target along x and y directions.

Dynamics of each robot arm is given by [41]

$$\dot{q}_{i1} = q_{i2} M_i(q_{i1}) \dot{q}_{i2} + V_i(q_{i1}, q_{i2}) q_{i2} + G_i(q_{i1}) + \tau_{id} = \tau_i \quad (80)$$

$i = 1, 2, \dots, 6$

where the vector $q_{i1} = [q_{i11}, q_{i12}]^T \in \mathbb{R}^2$ and $q_{i2} = [q_{i21}, q_{i22}]^T \in \mathbb{R}^2$ represent the position and velocity in x and y directions, respectively. The dynamic of two-link robot arm can be described as (80), where inertia matrix $M_i(q_{i1})$, centripetal and Coriolis matrix $V_i(q_{i1}, q_{i2})$ and gravitational

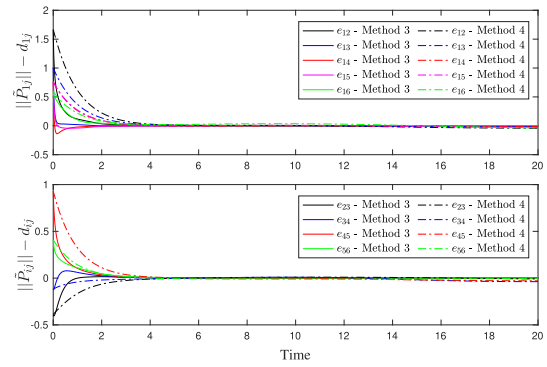


Fig. 15. Distance errors e_{ij} , $(i, j) \in E$ of the proposed method in Theorem 2 (Method 3) and the modified displacement-based method (Method 4) [26].

vector $G_i(q_{i1})$ are defined as follows [40]:

$$M_i(q_{i1}) = \begin{bmatrix} M_{i11} & M_{i12} \\ M_{i21} & M_{i22} \end{bmatrix} \in \mathbb{R}^{2 \times 2} \quad (81)$$

where

$$M_{i11} = (m_{i1} + m_{i2})r_{i1}^2 + 2m_{i2}r_{i1}r_{i2}\cos(q_{i12})$$

$$M_{i12} = M_{i21} = m_{i2}r_{i1}^2 + m_{i2}r_{i1}r_{i2}\cos(q_{i12})$$

$$M_{i22} = m_{i12}r_{i12}^2.$$

$$V_i(q_{i1}, q_{i2}) = \begin{bmatrix} V_{i11} & V_{i12} \\ V_{i21} & V_{i22} \end{bmatrix} \in \mathbb{R}^{2 \times 2} \quad (82)$$

where

$$V_{i11} = -m_{i2}r_{i1}r_{i2}\sin(q_{i12})q_{i22}$$

$$V_{i12} = -m_{i2}r_{i1}r_{i2}\sin(q_{i12})q_{i22} - m_{i2}r_{i1}r_{i2}\sin(q_{i12})q_{i21}$$

$$V_{i21} = m_{i2}r_{i1}r_{i2}\sin(q_{i12})q_{i22}$$

$$V_{i22} = 0.$$

$$G_i(q_{i1}, q_{i2}) = (G_{i11}, G_{i12})^T \in \mathbb{R}^2, \quad (83)$$

where $G_{i11} = (m_{i1} + m_{i2})gr_{i1}\cos(q_{i11}) + m_{i2}gr_{i2}\cos(q_{i11} + q_{i22})$, and $G_{i12} = m_{i2}gr_{i2}\cos(q_{i21} + q_{i12})$. In (80), the variable τ_{id} represents a constant bounded disturbance in

i	1	2	3	4	5	6
q_{i11}	0	2.5	2	0.7	-1.7	-1.9
q_{i12}	2.1	0	-1.1	-1.8	-1	1
q_{i21}	0.3	0.8	0	-0.5	0.5	0.4
q_{i22}	0.3	0	0.7	-1	1	-0.5

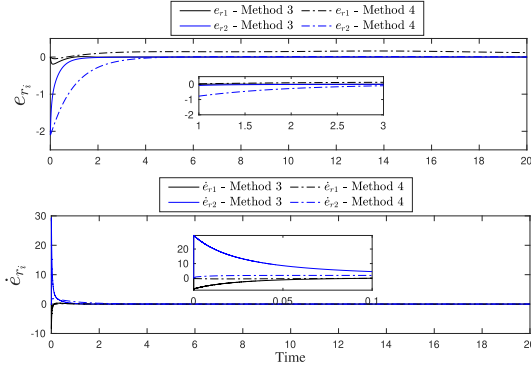


Fig. 16. Target tracking error of the leader and its time-derivative of the proposed method in Theorem 2 (Method 3) and the modified displacement-based method (Method 4) [26] along x and y directions.

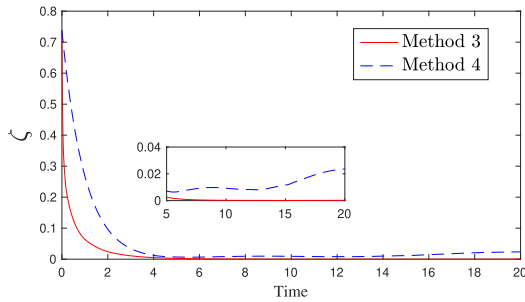


Fig. 17. Performance comparison (AFDE) between the proposed method in Theorem 2 (Method 3) and the modified displacement-based method (Method 4) [26].

the dynamics of each two-arm robot. The parameters of the each robot arm are $g = 9.8 \text{ m/s}^2$, $r_{i1} = 1 \text{ m}$, $r_{i2} = 1 \text{ m}$, $m_{i1} = 1 \text{ kg}$, and $m_{i2} = 1 \text{ kg}$, $\tau_{id} = [0.1, 0.1]^T$ ($i \in \{1, \dots, 6\}$). The initial conditions of all agents are given in Table II. The target velocity and initial position are $v_r = [\frac{1}{2\pi} \cos(\frac{1.5\pi}{20} t), \frac{1}{2\pi} \sin(\frac{1.5\pi}{20} t)]^T$ and $p_r(0) = [0, 0]^T$, respectively. We selected $b = 2$ and $k_r = 1$.

We applied an RBFNN with 81 neurons and activation function centers, μ_j , evenly spaced for each agent. One can obtain $g_i = 1/[(m_{i1} + m_{i2})r_{i1}^2 + 2m_{i2}r_{i2}^2 + 3m_{i2}r_{i1}r_{i2}]$ for $\forall i \in \{1, \dots, 6\}$ [40]. The values of the constants are $\kappa = 0.2$, $\Pi_i = 1$, $c = 20$, $k_v = 0.3$ and the desired distance is $d = 1.6$ for all edges. Figs. 12–16 show the simulation results of the proposed control law (31) applied to a nonlinear multiagent system. Fig. 12 indicates how these agents form an equilateral hexagon, where the desired distance for each side is $d = 1.6$. Fig. 13 represents the trajectories of each agent and the target in x and y directions. Velocity of each agent and the target in x and y directions are shown in Fig. 14.

To compare the results with the displacement-based method, the modified version of [26] has been simulated. By applying this method and adding constant displacements of the desired positions [35, p. 127] to the formation control problem, we obtain a shape-based control with distance errors shown in Fig. 15. Also, target tracking errors of the leader along x and y coordinates and time-derivatives for both methods are shown in Fig. 16. We use AFDE to compare our results with [26], as shown in Fig. 17. It can be seen that the proposed control method improves the AFDE when applied to an input-affine, nonlinear multiagent system.

V. CONCLUSION

This article proposed a control method for distance-based formation control and target tracking for nonlinear multiagent systems with unknown dynamics and disturbance. We considered second-order nonlinear agents that are connected through an undirected graph network. An RBFNN compensates for the agent's unknown dynamics. The proposed control law is based on the target tracking backstepping controller and is paired with a stable NN tuning law that ensures the stability of the obtained formation. The stability results show that distance error dynamics and NN weights' norm estimation error are UUB. Simulation results are provided to illustrate the performance and effectiveness of the proposed method. The performance index has been introduced to quantify the comparison with two existing displacement control methods. We demonstrated that the proposed method improves the system performance compared with two other displacement-based methods.

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